Access url:
https://docs.google.com/document/d/1CHT2tr4Czc49WzYsxUwqWE9uR-Bxwk6rqCw-JTFJcWs/edit?usp=sharing

1. Find the value of T(2) for the recurrence relation T(n) = 3T(n-1) + 12n, given that T(0)=5.

Given:

$T(n) = 3T(n-1) + 12n$

$T(0) = 5$

$\therefore\ T(1) = 3T(0) + 12(1)$

$T(1) = 3(5) + 12$

$= 27$

$\therefore\ T(2) = 3T(1) + 12(2)$

$= 3.(27) + 24$

$= 81 + 24$

$T(2) = 105$ (Ans)

2. Given a recurrence relation, solve it using the substitution method:

a. T(n) = T(n-1) + c

Given:

$$T(n) = \begin{cases} 1 & \text{if } n <= 1 \\ T(n-1)+c & \text{if } n > 1 \end{cases}$$

$$T(n) = T(n-1) + c$$

$$\therefore T(n-1) = T(n-1-1) + c$$

$$= T(n-2) + c + c$$

$$T(n-2) = T((n-2)-1) + c + c$$

$$= T(n-3) + c + c + c$$

$$= T(n-3) + 3c$$

$\vdots$ k steps

$$T(n-k) = T(n-k+1) + kC$$

let $n-k = 1 \Rightarrow k = n-1$

$\therefore$

$$\therefore \overset{k=n-1}{T(k)} = T(n-(n-1)+1) + kC$$

$$= T(0) + kC$$

$$= 1 + kC$$

$$= 1 + (n-1)C$$

$$= nC - C + 1$$

$$= O(n)$$

b. T(n) = 2T(n/2) + n

Given:

$T(n) = 2T(n/2) + n \;;\quad n > 1$

$T(n) = 1 \;;\quad n == 1$

$\therefore T(n/2) = 2T\left(\frac{n}{4}\right) + \frac{n}{2} \quad \text{----} (1)$

$T(n) = 2\left[2T(n/2^2) + \frac{n}{2}\right] + n$

$= 2^2 T\left(\frac{n}{2^2}\right) + n + n \quad = 2^2\left[2T(n/2^3) + \frac{n}{2^2}\right] + n + n$

$\therefore T(n/2^2) = 2T(n/2^3) + \frac{n}{2^2}$

$= 2^3 T(n/2^3) + n + n + n$

$\vdots$ k times

$= 2^k T(n/2^k) + k.N$

when $\frac{n}{2^k} = 1$

$\Rightarrow n = 2^k$

$\Rightarrow k = \log_2 n$

$= 2^k T(1) + k.N$

$= 2^k * 1 + k.N$

$= n + n.\log n$

$= O(n \log n)$

c. T(n) = 2T(n/2) + c

Given:

$$T(n) = \begin{cases} 1 & n == 1 \\ 2T(n/2) + c & n > 1 \end{cases}$$

$$T\left(\frac{n}{2}\right) = 2T\left(\frac{n}{4}\right) + c$$

$$T\left(\frac{n}{4}\right) = 2T\left(\frac{n}{2^3}\right) + c$$

$$\therefore T(n) = 2\left[2T\left(\frac{n}{4}\right) + c\right] + c$$

$$= 2^2 T\left(\frac{n}{2^2}\right) + 2c$$

$$= 2^3 T\left(\frac{n}{2^3}\right) + 3c$$

$\vdots$ k times

$$= 2^k T\left(\frac{n}{2^k}\right) + kC$$

$$= n.\, T\left(\frac{n}{n}\right) + k.C$$

$$= n.\, T(1) + k.C$$

$$= n + k.C$$

$$= O(n + \log n) \text{ (Ans)}$$

$n = 2^k$

$\therefore \log n = \log 2^k$

$\therefore \log n = k.1$

$\Rightarrow k = \log n$

d. T(n) = T(n/2) + c

Given:

$$T(n) = \begin{cases} 1 & n == 1 \\ T(n/2) + c & n > 1 \end{cases}$$

$T(n) = T(n/2) + c$

$T\left(\frac{n}{2}\right) = T\left(\frac{n}{4}\right) + c$

$T\left(\frac{n}{4}\right) = T\left(\frac{n}{8}\right) + c$

$$T(n) = T\left(\frac{n}{2}\right) + c$$

$$= T\left(\frac{n}{2^2}\right) + c + c$$

$$= T\left(\frac{n}{2^3}\right) + 3c$$

$\vdots$ k times

$$= T\left(\frac{n}{2^k}\right) + kc$$

let $n = 2^k$

$\therefore \log n = \log 2^k$

$\Rightarrow \log n = k \cdot 1$

$$= T(n/n) + \log n \cdot c$$

$$= 1 + \log n \cdot c$$

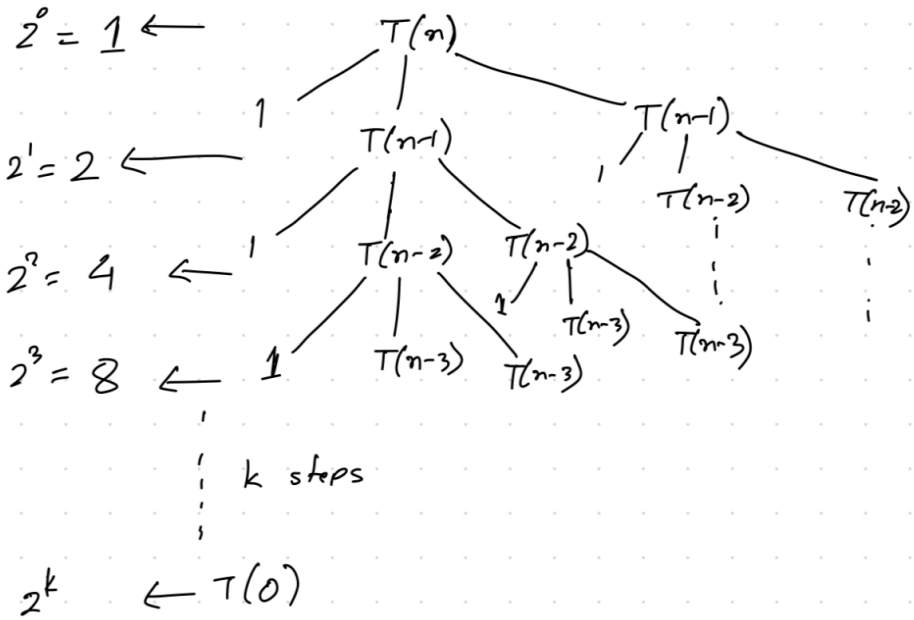
$$= O(\log n)$$ (Ans)

3. Given a rec%rrence relation, solve it %sing the rec%rsive tree approach:

a. T(n) = 2T(n-1) +1

Given :

$$T(n) = \begin{cases} 1 & n = 0 \\ 2T(n-1) + 1 & n > 0 \end{cases}$$



$$1 + 2 + 4 + 8 + \cdots + 2^k = 2^{k+1} - 1$$

Assume $n - k = 0$
$\Rightarrow k = n$

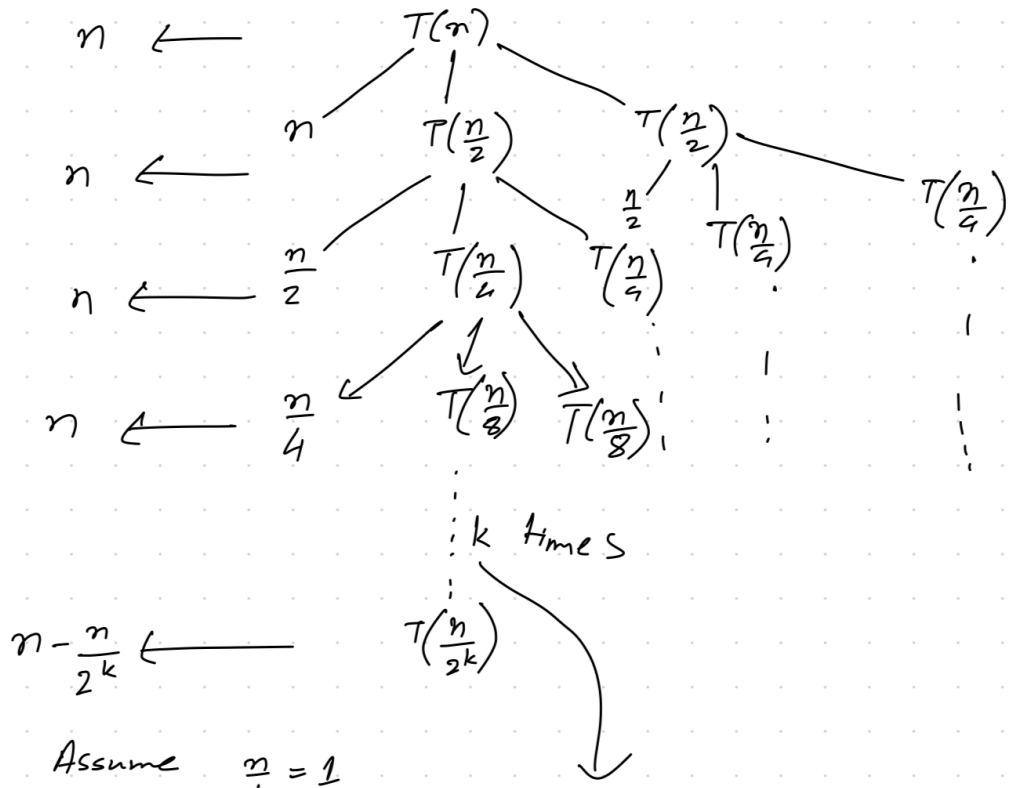
$$= 2^{n+1} - 1$$

$$= \Theta(2^n) \quad \text{(Ans)}$$

b. T(n) = 2T(n/2) + n

Given:

$$T(n) = \begin{cases} 1 & n == 1 \\ 2T\left(\frac{n}{2}\right) + n & n > 1 \end{cases}$$



Assume $\frac{n}{2^k} = 1$

$\therefore n = 2^k$

$\Rightarrow \log n = \log 2^k$

$\Rightarrow \log n = k \cdot 1$

$\Rightarrow k = \log n$

$T(n) = n + n + n + \cdots + n$

$= n(1 + 1 + 1 + \cdots + 1)$

$= O(n \log(n))$ (Ans)